



Faculté des Sciences et Technologies (FST)

Rapport du travail de Laboratoire N° 2_ Administration Système

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Niveau : L3

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L'objectif de ce TD est de :

- Comprendre le principe d'une sauvegarde
- Utiliser tar et rsync
- Automatiser une sauvegarde avec cron
- Vérifier et restaurer une sauvegarde
- Mettre en place une sauvegarde vers un serveur distant
- Utiliser rsync via SSH
- Sécuriser la connexion par clé SSH
- Restaurer les données

Exercice 1

```
donsam@donsamserver:~$ sudo mkdir /backup
donsam@donsamserver:~$ sudo mkdir -p /var/www/
donsam@donsamserver:~$ sudo mkdir -p /etc/
donsam@donsamserver:~$ sudo mkdir -p /home/devops/projects
donsam@donsamserver:~$ cd /home/devops/projects
donsam@donsamserver:/home/devops/projects$ ls
donsam@donsamserver:/home/devops/projects$ cd
donsam@donsamserver:~$ sudo touch /home/devops/projects/fichier1.txt
donsam@donsamserver:~$ sudo touch /home/devops/projects/fichier2.txt
```

```
donsam@donsamserver:~$ sudo nano /home/devops/projects/fichier1.txt
donsam@donsamserver:~$ cat /home/devops/projects/fichier1.txt
TP de sauvegardes locales
```

```
donsam@donsamserver:~$ sudo tar -czf /backup/data_backup.tar.gz /home/de
vops/projects
tar: Removing leading '/' from member names
donsam@donsamserver:~$ sudo tar -tzf /backup/data_backup.tar.gz
home/devops/projects/
home/devops/projects/fichier2.txt
home/devops/projects/fichier1.txt
donsam@donsamserver:~$ |
```

```
donsam@donsamserver:~$ sudo apt install rsync -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
rsync est déjà la version la plus récente (3.2.7-lubuntu1.2).
rsync passé en « installé manuellement ».
0 mis à jour, 0 nouvellement installés, 0 à enlever et 28 non mis à jour
.
donsam@donsamserver:~$ rsync -av /home/devops/projects /backup/
sending incremental file list
rsync: [generator] recv_generator: mkdir "/backup/projects" failed: Perm
ission denied (13)
*** Skipping any contents from this failed directory ***
projects/

sent 127 bytes  received 20 bytes  294,00 bytes/sec
total size is 26  speedup is 0,18
rsync error: some files/attrs were not transferred (see previous errors)
(code 23) at main.c(1338) [sender=3.2.7]
donsam@donsamserver:~$ nano sauvegarde.sh
donsam@donsamserver:~$ chmod +x sauvegarde.sh
donsam@donsamserver:~$ ./sauvegarde.sh
tar: Removing leading '/' from member names
tar (child): /backup/data-2026-03-16-00-14.tar.gz: Cannot open: Permissi
on denied
tar (child): Error is not recoverable: exiting now
tar: Child returned status 2
tar: Error is not recoverable: exiting now
donsam@donsamserver:~$ sudo ./sauvegarde.sh
tar: Removing leading '/' from member names
donsam@donsamserver:~$ |
```

```
donsam@donsamserver:~$ crontab -e
no crontab for donsam - using an empty one

Select an editor. To change later, run 'select-editor'.
 1. /bin/nano          <---- easiest
 2. /usr/bin/vim.basic
 3. /usr/bin/vim.tiny
 4. /bin/ed

Choose 1-4 [1]: 1
crontab: installing new crontab
donsam@donsamserver:~$ ls /backup
data-2026-03-16-00-14.tar.gz  data_backup.tar.gz
donsam@donsamserver:~$ |
```

```
donsam@donsamserver:~$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirect
ed).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cr
on(8)
#
# m h dom mon dow  command
* * * * * /home/devops/projects/sauvegarde.sh
donsam@donsamserver:~$ ls /backup
data-2026-03-16-00-14.tar.gz  data_backup.tar.gz
donsam@donsamserver:~$ |
```

Exercise 2

```
donsam@donsamserver:~$ sudo adduser backupuser
info: Adding user `backupuser' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `backupuser' (1001) ...
info: Adding new user `backupuser' (1001) with group `backupuser
(1001)' ...
info: Creating home directory `/home/backupuser' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for backupuser
Enter the new value, or press ENTER for the default
  Full Name []: Admin Gabard
  Room Number []: 01
  Work Phone []: +509 34545832
  Home Phone []:
  Other []:
Is the information correct? [Y/n] Y
info: Adding new user `backupuser' to supplemental / extra group
s `users' ...
info: Adding user `backupuser' to group `users' ...
donsam@donsamserver:~$
```

```
donsam@donsamserver:~$ sudo mkdir /backup/devcloud
donsam@donsamserver:~$ sudo chmod 777 /backup/devcloud
donsam@donsamserver:~$ sudo mkdir -p /var/www/
donsam@donsamserver:~$ sudo mkdir -p /etc/
donsam@donsamserver:~$ sudo mkdir -p /home/devops/projects/
donsam@donsamserver:~$ sudo chown backupuser:backupuser /backup/
devcloud
donsam@donsamserver:~$
```

```

donsam@donsamserver:~$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/donsam/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/donsam/.ssh/id_rsa
Your public key has been saved in /home/donsam/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:sFrG7bHp2M0Q90w5EdN5HyCvgo7Z0Uz5IQFEfI1vhRo donsam@donsam
server
The key's randomart image is:
+---[RSA 4096]-----+
|      ++..o...      |
|      . Eo+oo.      |
|      . o+*.+. ...  |
|      . ==+o=o. . . |
|      =oS+=o       . |
|      +=..*.o       |
|      .o o= +       |
|      + + .         |
|      . o o         |
+---[SHA256]-----+

```

```

donsam@donsamserver:~$ cat .ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQADokAn5NFSoeYPB01666iQwQIBc
L8ktgK8cWoGdZ3VWtYSdetRdPzTsxPMSSayWNXTvZLorvnbx6T2wqgS6yhYqDlou
XF67xbqMvFmEVUzB0ru7jmjdnCaZzE10Uslvg0VdlvpmY5Bw0GYxmZmDLgB+9GVS
AXK4vdlw10PDXEzozqo4vxRUSGPzqjuje990S5Hp6UKPLyKMFC/ndT3zgUDW0+S/
YJhpE/JNVL0Q0re/nP/Y7Q0DbTLev+CHGbLDXD/WnObdc0ozFvv+8DnuXkZ0Hlko
PIPT7CiqI4SAFHUP/YtjLvTfMdbCoD73aMniMTRfPzwX09vXp8dWwEqtPhJ20dnj
b4kCpzM+Eb7AS0BUwuCm1vuLLqPOIv4eyI8ciytKuWFwqf3nukwAt4+QYdidoZ4b
GeRG1TSPpaA2oNklk92Y4hEgg2hymZa9pEkS5a2Icsd2zJ1505KtoN/jdFepeRWn
Uq1t76FLL+kF08FskkGWEfUipxpgb5RnMwu+r9potSx/vmxqlXxE6gdh5Tuywe6/
DB89qgEZ2hdhnb8P/oKrDRn+PMrbgX58rtqX+cyLrvdl0ji4TCsb14pILQZxf8P2
CikGqYi1c2hRy2cwh3TY2JBoWiDHTZPEiwy94jT2ySY7pupJDWuNXRiE4uxDcAwf
LR3Gn4RVBLa8mZ6Qfw== donsam@donsamserver

```

```

donsam@donsamserver:~$ ssh-copy-id donsam@192.168.56.109
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/
home/donsam/.ssh/id_rsa.pub"
The authenticity of host '192.168.56.109 (192.168.56.109)' can't
be established.
ED25519 key fingerprint is SHA256:uUDXgG0lH5U5gtMON68Q1assy4VkuH
9K9uPm0Q431uA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprin
t])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new ke
y(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- i
f you are prompted now it is to install the new keys
donsam@192.168.56.109's password:
Permission denied, please try again.
donsam@192.168.56.109's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'donsam@192.168.5
6.109'"
and check to make sure that only the key(s) you wanted were adde
d.

donsam@donsamserver:~$ |

```

```

donsam@donsamserver:~$ ls /home/devops
projects
donsam@donsamserver:~$ sudo rsync -avz backupuser@192.168.56.109
:/backup/dev/cloud
The authenticity of host '192.168.56.109 (192.168.56.109)' can't
be established.
ED25519 key fingerprint is SHA256:uUDXgG0lH5U5gtMON68Q1assy4VkuH
9K9uPm0Q431uA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprin
t])? yes
Warning: Permanently added '192.168.56.109' (ED25519) to the lis
t of known hosts.
backupuser@192.168.56.109's password:
receiving incremental file list
rsync: [sender] change_dir "/backup/dev" failed: No such file or
directory (2)

sent 8 bytes  received 8 bytes  0,52 bytes/sec
total size is 0  speedup is 0,00
rsync error: some files/attrs were not transferred (see previous
errors) (code 23) at main.c(1865) [Receiver=3.2.7]

```

```
donsam@donsamserver:~$ nano sauvegarde_distance.sh
donsam@donsamserver:~$ chmod +x sauvegarde_distance.sh
donsam@donsamserver:~$ sudo ./sauvegarde_distance.sh
[sudo] password for donsam:
backupuser@192.168.56.109's password:
donsam@donsamserver:~$ crontab -e
crontab: installing new crontab
donsam@donsamserver:~$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
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#
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# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
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ed).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cr
on(8)
#
# m h dom mon dow  command
```

```
donsam@donsamserver:~$ sudo rsync -avz backupuser@192.168.56.109
:/backup/devcloud /home/devops
backupuser@192.168.56.109's password:
receiving incremental file list
devcloud/
devcloud/.pwd.lock
devcloud/. resolv.conf.systemd-resolved.bak
devcloud/.updated
devcloud/adduser.conf
devcloud/bash.bashrc
devcloud/bash_completion
devcloud/bindresvport.blacklist
devcloud/ca-certificates.conf
devcloud/crontab
devcloud/crypttab
devcloud/debconf.conf
devcloud/debian_version
devcloud/deluser.conf
devcloud/dhcpd.conf
devcloud/e2scrub.conf
devcloud/environment
devcloud/ethertypes
devcloud/fstab
devcloud/fuse.conf
devcloud/gai.conf
devcloud/group
devcloud/group-
devcloud/gshadow
devcloud/gshadow-
devcloud/hdparm.conf
devcloud/host.conf
devcloud/hostname
devcloud/hosts
```

```
devcloud/xml/polkitd.xml
devcloud/xml/polkitd.xml.old
devcloud/xml/xml-core.xml
devcloud/xml/xml-core.xml.old

sent 19.403 bytes  received 656.876 bytes  104.042,92 bytes/sec
total size is 2.461.498  speedup is 3,64
donsam@donsamserver:~$ ls /home/devops
devcloud  projects
donsam@donsamserver:~$ |
```

Questions

1. Quelle est la différence entre tar et rsync ?

Rep : tar est un outil conçu pour archiver des fichiers et dossiers en un seul fichier, souvent compressé. Il est particulièrement utile pour créer des sauvegardes complètes et cohérentes, car il préserve les métadonnées (permissions, propriétaires, timestamps, etc.).

rsync, en revanche, est un outil de synchronisation de fichiers et dossiers entre deux emplacements, qu'ils soient locaux ou distants. Contrairement à tar, rsync ne crée pas d'archive unique, mais copie uniquement les fichiers modifiés ou ajoutés depuis la dernière synchronisation. Cela le rend extrêmement efficace pour les sauvegardes incrémentielles.

2. Différence entre scp et rsync ?

Rep : scp (Secure Copy Protocol) est un outil simple et sécurisé pour copier des fichiers entre deux machines via SSH. Il est intégré à la suite OpenSSH et utilise le chiffrement pour protéger les données pendant le transfert.

rsync, comme mentionné précédemment, est bien plus intelligent : il ne copie que les modifications depuis la dernière synchronisation, ce qui économise énormément de bande passante et de temps.

3. Pourquoi automatiser les sauvegardes ?

Rep : Automatiser les sauvegardes est une bonne pratique essentielle pour plusieurs raisons car l'automatisation des sauvegardes élimine les risques liés aux oublis ou erreurs humaines, garantissant une protection fiable et constante des données. Grâce à une planification précise (quotidienne, hebdomadaire, etc.), elle assure que toutes les modifications récentes sont systématiquement sauvegardées, sans dépendre d'une intervention manuelle. Ce processus libère les administrateurs des tâches répétitives, tout en offrant une restauration rapide en cas de panne, suppression accidentelle ou cyberattaque, réduisant ainsi les temps d'arrêt. Enfin, l'automatisation standardise les procédures, assurant une consistance optimale et minimisant les risques d'omissions ou d'incohérences.

4. Où sont stockées les tâches cron de l'utilisateur ?

Rep : Les tâches cron sont stockées dans le fichier crontab de l'utilisateur actuel.

5. Comment planifier une sauvegarde tous les jours à 23h ?

Rep : 0 23 * * * /chemin/vers/sauvegarde.sh

Conclusion

Avec ce TD, j'ai compris comment automatiser mes sauvegardes pour éviter les oublis, gagner du temps et protéger mes données en cas de panne ou d'attaque. Grâce à cron et rsync, je peux maintenant planifier des sauvegardes fiables et restaurer mes fichiers rapidement si nécessaire.